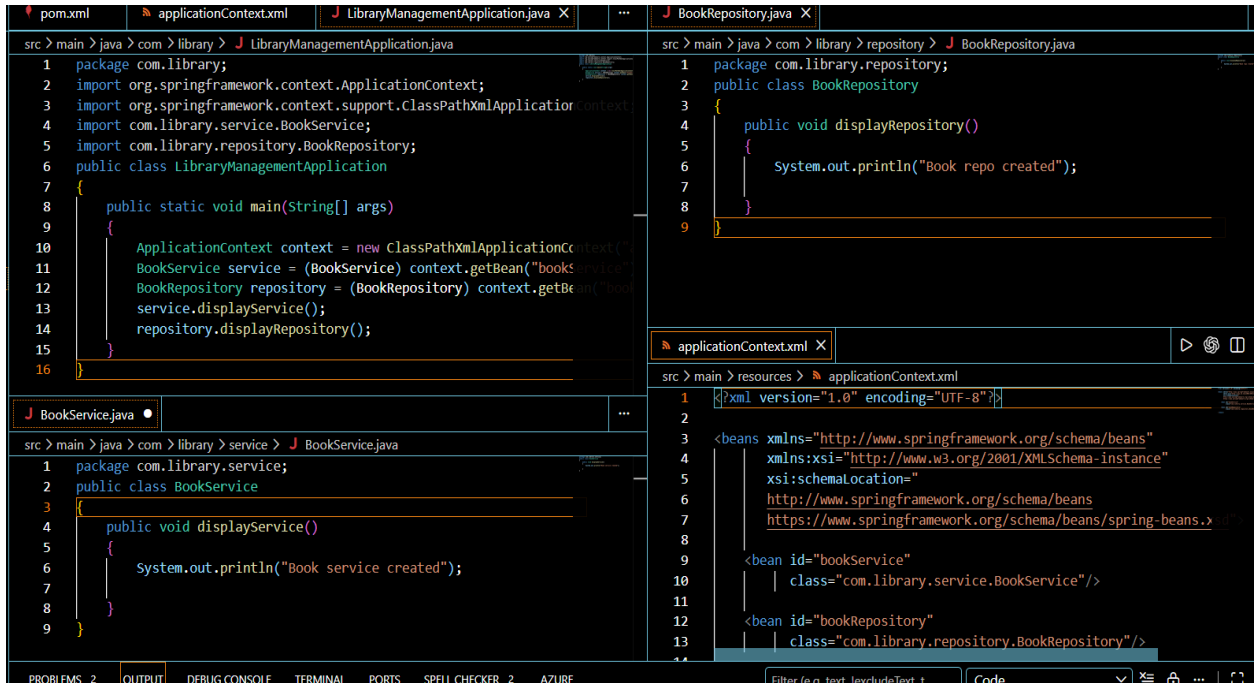


Exercise 1: Configuring a Basic Spring Application



```
pom.xml
src > main > java > com > library > LibraryManagementApplication.java
1 package com.library;
2 import org.springframework.context.ApplicationContext;
3 import org.springframework.context.support.ClassPathXmlApplicationContext;
4 import com.library.service.BookService;
5 import com.library.repository.BookRepository;
6 public class LibraryManagementApplication
7 {
8     public static void main(String[] args)
9     {
10         ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml");
11         BookService service = (BookService) context.getBean("bookService");
12         BookRepository repository = (BookRepository) context.getBean("bookRepository");
13         service.displayService();
14         repository.displayRepository();
15     }
16 }

applicationContext.xml
src > main > resources > applicationContext.xml
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5       xsi:schemaLocation="
6         http://www.springframework.org/schema/beans
7         https://www.springframework.org/schema/beans/spring-beans.xsd" >
8
9     <bean id="bookService"
10          class="com.library.service.BookService"/>
11
12     <bean id="bookRepository"
13          class="com.library.repository.BookRepository"/>
14
15 </beans>

LibraryManagementApplication.java
src > main > java > com > library > LibraryManagementApplication.java
1 package com.library;
2 import org.springframework.context.ApplicationContext;
3 import org.springframework.context.support.ClassPathXmlApplicationContext;
4 import com.library.service.BookService;
5 import com.library.repository.BookRepository;
6 public class LibraryManagementApplication
7 {
8     public static void main(String[] args)
9     {
10         ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml");
11         BookService service = (BookService) context.getBean("bookService");
12         BookRepository repository = (BookRepository) context.getBean("bookRepository");
13         service.displayService();
14         repository.displayRepository();
15     }
16 }

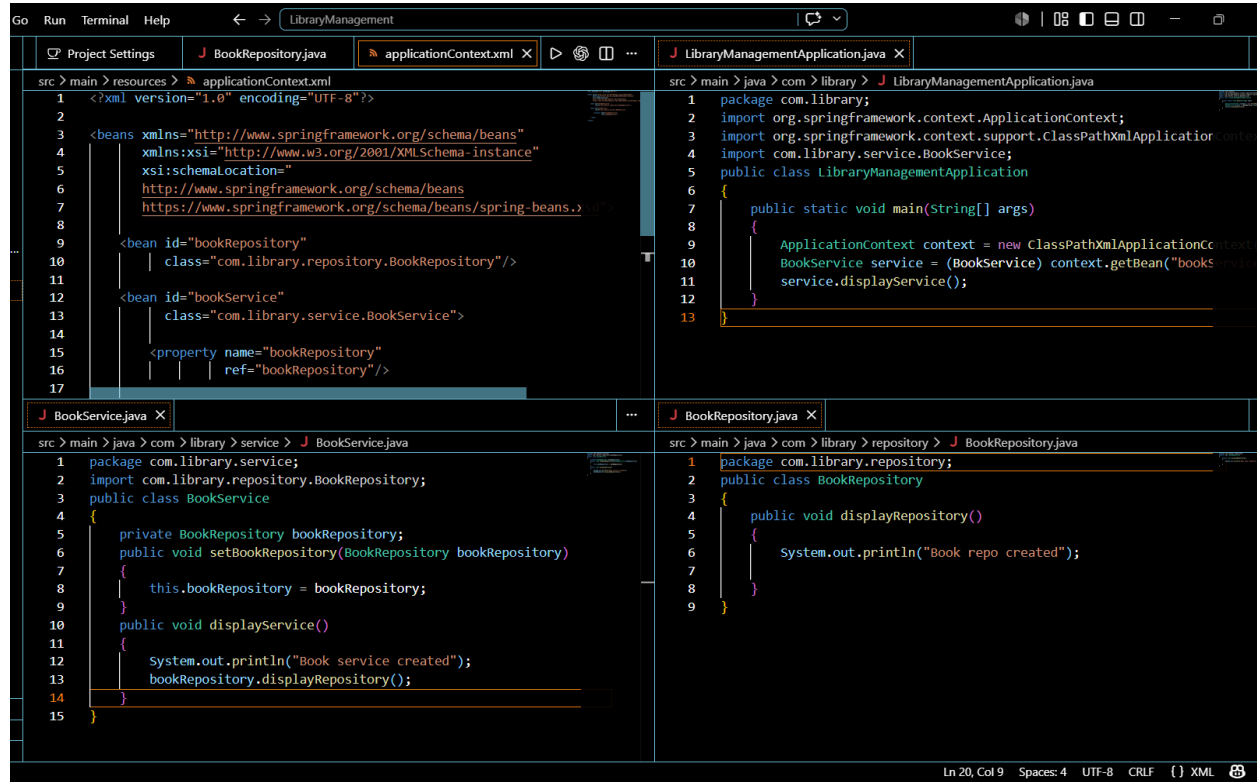
BookRepository.java
src > main > java > com > library > repository > BookRepository.java
1 package com.library.repository;
2 public class BookRepository
3 {
4     public void displayRepository()
5     {
6         System.out.println("Book repo created");
7     }
8 }

BookService.java
src > main > java > com > library > service > BookService.java
1 package com.library.service;
2 public class BookService
3 {
4     public void displayService()
5     {
6         System.out.println("Book service created");
7     }
8 }

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 2 AZURE
```

```
INFO] -----< com.library:LibraryManagement >-----
--
S C:\Users\USER\Downloads\cognizant\week2\LibraryManagement> mvn exec:java
-Dexec.mainClass=com.library.LibraryManagementApplication"
INFO] Scanning for projects...
INFO]
INFO] -----< com.library:LibraryManagement >-----
--
INFO] Building LibraryManagement 1.0
INFO]   from pom.xml
INFO] -----[ jar ]-----
--
INFO]
INFO] --- exec:3.1.0:java (default-cli) @ LibraryManagement ---
Book service created
Book repo created
INFO] -----
--
INFO] BUILD SUCCESS
INFO] -----
--
INFO] Total time: 0.452 s
INFO] Finished at: 2026-06-23T22:24:30+05:30
INFO] -----
S C:\Users\USER\Downloads\cognizant\week2\LibraryManagement> |
```

Exercise 2: Implementing Dependency Injection



```
src > main > resources > applicationContext.xml
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5       xsi:schemaLocation="
6         http://www.springframework.org/schema/beans
7         https://www.springframework.org/schema/beans/spring-beans.xsd"
8       >
9   <bean id="bookRepository"
10        class="com.library.repository.BookRepository"/>
11
12   <bean id="bookService"
13        class="com.library.service.BookService">
14
15     <property name="bookRepository"
16              ref="bookRepository"/>
17
18 </beans>

src > main > java > com > library > LibraryManagementApplication.java
1 package com.library;
2 import org.springframework.context.ApplicationContext;
3 import org.springframework.context.support.ClassPathXmlApplicationContext;
4 import com.library.service.BookService;
5 public class LibraryManagementApplication
6 {
7     public static void main(String[] args)
8     {
9         ApplicationContext context = new ClassPathXmlApplicationContext(
10             "applicationContext.xml");
11         BookService service = (BookService) context.getBean("bookService");
12         service.displayService();
13     }
14 }

src > main > java > com > library > service > BookService.java
1 package com.library.service;
2 import com.library.repository.BookRepository;
3 public class BookService
4 {
5     private BookRepository bookRepository;
6     public void setBookRepository(BookRepository bookRepository)
7     {
8         this.bookRepository = bookRepository;
9     }
10    public void displayService()
11    {
12        System.out.println("Book service created");
13        bookRepository.displayRepository();
14    }
15 }

src > main > java > com > library > repository > BookRepository.java
1 package com.library.repository;
2 public class BookRepository
3 {
4     public void displayRepository()
5     {
6         System.out.println("Book repo created");
7     }
8 }
9 }
```

```
PS C:\Users\USER\Downloads\cognizant\week2\LibraryManagement> mvn exec:java
-Dexec.mainClass=com.library.LibraryManagementApplication"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.library:LibraryManagement >-----
[INFO]
[INFO] Building LibraryManagement 1.0
[INFO]   from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec:3.1.0:java (default-cli) @ LibraryManagement ---
Book service created
Book repo created
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.800 s
[INFO] Finished at: 2026-06-23T21:44:24+05:30
[INFO] -----
PS C:\Users\USER\Downloads\cognizant\week2\LibraryManagement> |
```

Exercise 4: Creating and Configuring a Maven Project

```
pom.xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.library</groupId>
  <artifactId>LibraryManagement</artifactId>
  <version>1.0</version>
  <dependencies>
    <!-- Spring Context -->
    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-context</artifactId>
      <version>5.3.39</version>
    </dependency>

    <!-- Spring AOP -->
    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-aop</artifactId>
      <version>5.3.39</version>
    </dependency>

    <!-- Spring Web MVC -->
    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-webmvc</artifactId>
      <version>5.3.39</version>
    </dependency>
  </dependencies>

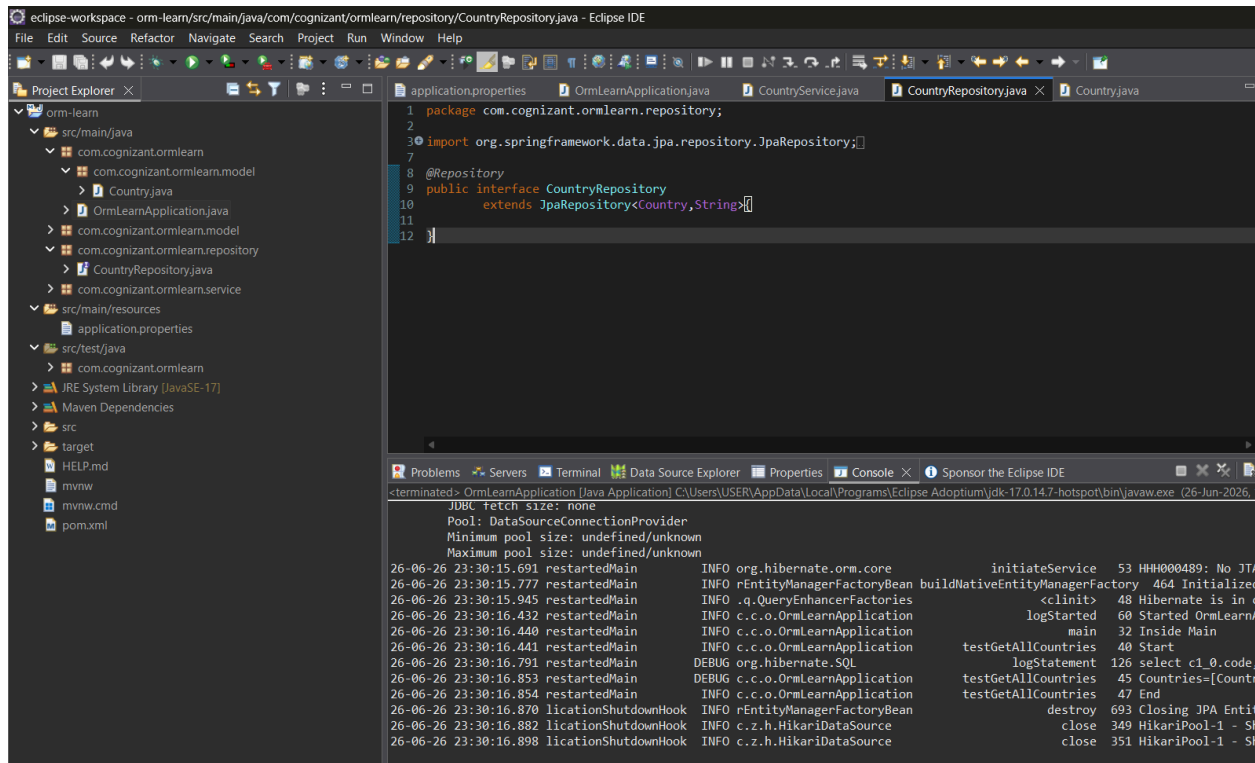
  <build>
    <plugins>

      <!-- Maven Compiler Plugin -->
      <plugin>
```

```

1.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\USER\Downloads\cognizant\week2\LibraryManagement\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ LibraryManagement ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ LibraryManagement ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ LibraryManagement ---
[INFO] Building jar: C:\Users\USER\Downloads\cognizant\week2\LibraryManagement\target\LibraryManagement-1.0.jar
[INFO]
[INFO] --- install:3.1.4:install (default-install) @ LibraryManagement ---
[INFO] Installing C:\Users\USER\Downloads\cognizant\week2\LibraryManagement\target\LibraryManagement-1.0.jar to C:\Users\USER\.m2\repository\com\library\LibraryManagement\1.0\LibraryManagement-1.0.jar
[INFO] Installing C:\Users\USER\Downloads\cognizant\week2\LibraryManagement\target\LibraryManagement-1.0.pom to C:\Users\USER\.m2\repository\com\library\LibraryManagement\1.0\LibraryManagement-1.0.pom
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 4.950 s
[INFO] Finished at: 2026-06-23T22:29:29+05:30
[INFO]
C:\Users\USER\Downloads\cognizant\week2\LibraryManagement> |
```

Spring Data JPA - Quick Example

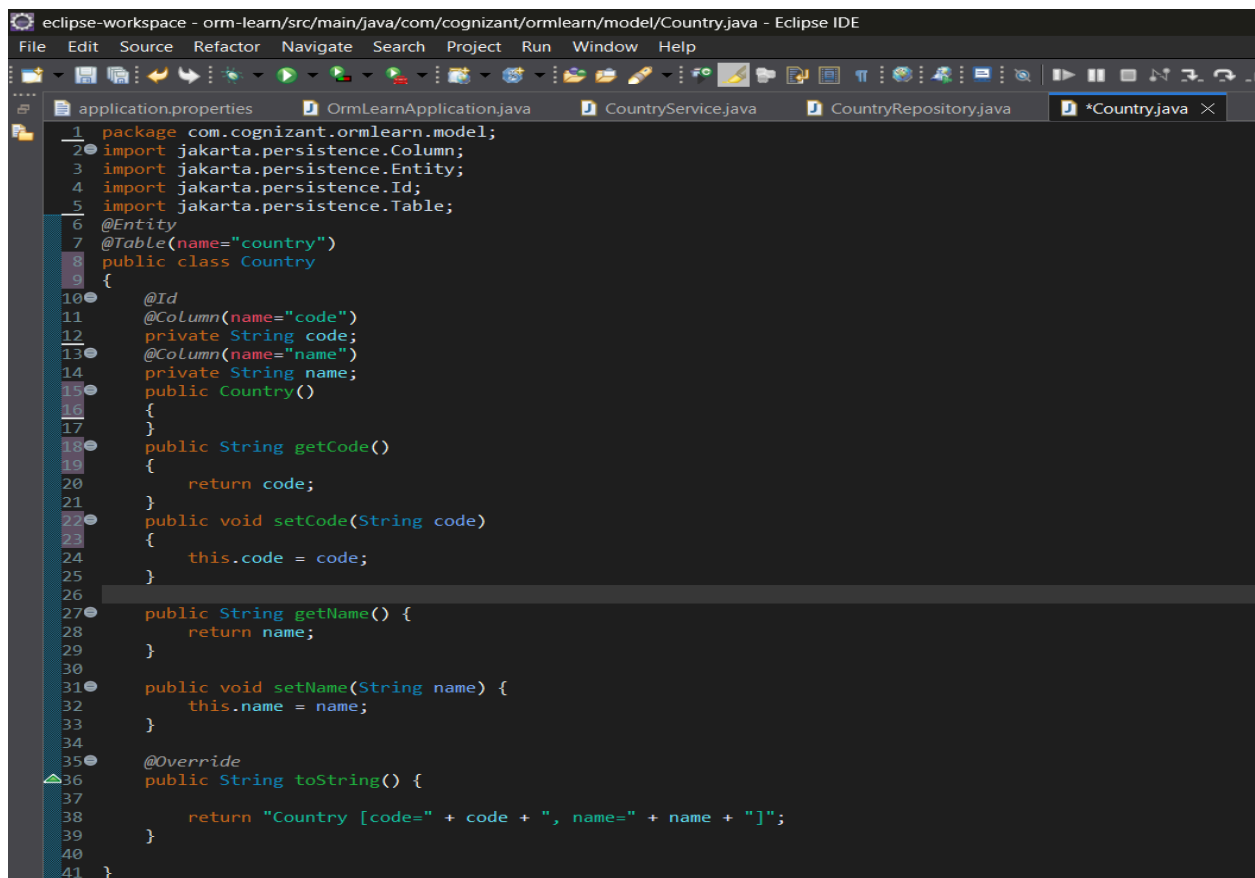


The screenshot shows the Eclipse IDE with the `CountryRepository.java` file open. The file is located in the package `com.cognizant.ormlearn.repository` and implements the `JpaRepository` interface. The console output shows the application running and logging the results of the `testGetAllCountries` method.

```
1 package com.cognizant.ormlearn.repository;
2
3 import org.springframework.data.jpa.repository.JpaRepository;
4
5 @Repository
6 public interface CountryRepository
7     extends JpaRepository<Country,String>{
8
9 }
10
11
12 }
```

Console Output:

```
<terminated> OrmLearnApplication [Java Application] C:\Users\USER\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.14-hotspot\bin\javaw.exe (26-Jun-2026,
JDBC fetch size: none
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
26-06-26 23:30:15.691 restartedMain INFO org.hibernate.orm.core initiateService 53 HHH000489: No JTA
26-06-26 23:30:15.777 restartedMain INFO rEntityManagerFactoryBean buildNativeEntityManagerFactory 464 Initialized
26-06-26 23:30:15.945 restartedMain INFO .q.QueryEnhancerFactories <clinit> 48 Hibernate is in c
26-06-26 23:30:16.432 restartedMain INFO c.c.o.OrmLearnApplication logStarted 60 Started OrmLearn
26-06-26 23:30:16.440 restartedMain INFO c.c.o.OrmLearnApplication main 32 Inside Main
26-06-26 23:30:16.441 restartedMain INFO c.c.o.OrmLearnApplication testGetAllCountries 40 Start
26-06-26 23:30:16.791 restartedMain DEBUG org.hibernate.SQL logStatement 126 select c1_0.code,
26-06-26 23:30:16.853 restartedMain DEBUG c.c.o.OrmLearnApplication testGetAllCountries 45 Countries=[Countr
26-06-26 23:30:16.854 restartedMain INFO c.c.o.OrmLearnApplication testGetAllCountries 47 End
26-06-26 23:30:16.870 licationShutdownHook INFO rEntityManagerFactoryBean destroy 693 Closing JPA Entit
26-06-26 23:30:16.882 licationShutdownHook INFO c.z.h.HikariDataSource close 349 HikariPool-1 - SH
26-06-26 23:30:16.898 licationShutdownHook INFO c.z.h.HikariDataSource close 351 HikariPool-1 - SH
```



The screenshot shows the Eclipse IDE with the `Country.java` file open. The file is located in the package `com.cognizant.ormlearn.model` and implements the `Entity` interface. The file defines the `Country` class with attributes `code` and `name`, and methods `getCode`, `setName`, and `toString`.

```
1 package com.cognizant.ormlearn.model;
2 import jakarta.persistence.Column;
3 import jakarta.persistence.Entity;
4 import jakarta.persistence.Id;
5 import jakarta.persistence.Table;
6 @Entity
7 @Table(name="country")
8 public class Country
9 {
10     @Id
11     @Column(name="code")
12     private String code;
13     @Column(name="name")
14     private String name;
15     public Country()
16     {
17     }
18     public String getCode()
19     {
20         return code;
21     }
22     public void setCode(String code)
23     {
24         this.code = code;
25     }
26
27     public String getName() {
28         return name;
29     }
30
31     public void setName(String name) {
32         this.name = name;
33     }
34
35     @Override
36     public String toString() {
37
38         return "Country [code=" + code + ", name=" + name + "];
39     }
40 }
41 }
```

eclipse-workspace - orm-learn/src/main/java/com/cognizant/ormlearn/OrmLearnApplication.java - Eclipse IDE

File Edit Source Refactor Navigate Search Project Run Window Help

application.properties *OrmLearnApplication.java CountryService.java CountryRepository.java

```
1 package com.cognizant.ormlearn;
2 import java.util.List;
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5 import org.springframework.boot.SpringApplication;
6 import org.springframework.boot.autoconfigure.SpringBootApplication;
7 import org.springframework.context.ApplicationContext;
8 import com.cognizant.ormlearn.model.Country;
9 import com.cognizant.ormlearn.service.CountryService;
10 @SpringBootApplication
11 public class OrmLearnApplication
12 {
13     private static final Logger LOGGER =LoggerFactory.getLogger(OrmLearnApplication.class);
14     private static CountryService countryService;
15     public static void main(String[] args)
16     {
17         ApplicationContext context =SpringApplication.run(OrmLearnApplication.class,args);
18         countryService =context.getBean(CountryService.class);
19         LOGGER.info("Inside Main");
20         testGetAllCountries();
21     }
22
23     private static void testGetAllCountries()
24     {
25
26         LOGGER.info("Start");
27
28         List<Country> countries =
29             countryService.getAllCountries();
30
31         LOGGER.debug("Countries={}",countries);
32
33         LOGGER.info("End");
34     }
35 }
36
```

```
123
124 • use ormlearn;
125 • CREATE TABLE country(
126     code VARCHAR(2) PRIMARY KEY,
127     name VARCHAR(50)
128 );
129
130 • INSERT INTO country VALUES ('IN','India');
131 • INSERT INTO country VALUES ('US','United States OF America');
```

Result Grid | Filter Rows: | Edit: | Export/Import

	code	name
▶	IN	India
	US	United States OF America
✱	NULL	NULL

Result Grid

```
eclipse-workspace - orm-learn/src/main/java/com/cognizant/ormlearn/service/CountryService

File Edit Source Refactor Navigate Search Project Run Window Help

application.properties *OrmLearnApplication.java CountryService.java

1 package com.cognizant.ormlearn.service;
2
3 import java.util.List;
4
5 import org.springframework.beans.factory.annotation.Autowired;
6 import org.springframework.stereotype.Service;
7
8 import jakarta.transaction.Transactional;
9
10 import com.cognizant.ormlearn.model.Country;
11 import com.cognizant.ormlearn.repository.CountryRepository;
12
13 @Service
14 public class CountryService {
15
16     @Autowired
17     CountryRepository countryRepository;
18
19     @Transactional
20     public List<Country> getAllCountries(){
21
22         return countryRepository.findAll();
23
24     }
25
26 }
```

```
<terminated> OrmLearnApplication [Java Application] C:\Users\USER\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.14.7-hotspot\bin\javaw.e

:: Spring Boot :: (v4.1.1.0)

26-06-26 23:38:06.008 restartedMain INFO c.c.o.OrmLearnApplication logStarting 54 S
26-06-26 23:38:06.012 restartedMain DEBUG c.c.o.OrmLearnApplication logStarting 55 R
26-06-26 23:38:06.013 restartedMain INFO c.c.o.OrmLearnApplication logStartupProfileInfo 652 N
26-06-26 23:38:06.085 restartedMain INFO ertyDefaultsPostProcessor log 69 D
26-06-26 23:38:06.671 restartedMain INFO toryConfigurationDelegate registerRepositoriesIn 145 B
26-06-26 23:38:06.771 restartedMain INFO toryConfigurationDelegate registerRepositoriesIn 213 F
26-06-26 23:38:07.246 restartedMain INFO org.hibernate.orm.jpa logPersistenceUnitInformation 1
26-06-26 23:38:07.338 restartedMain INFO org.hibernate.orm.core logVersion 41 H
26-06-26 23:38:07.994 restartedMain INFO SpringPersistenceUnitInfo addTransformer 129 N
26-06-26 23:38:08.038 restartedMain INFO c.z.h.HikariDataSource getConnection 109 H
26-06-26 23:38:08.476 restartedMain INFO c.z.h.pool.HikariPool checkFailFast 580 H
26-06-26 23:38:08.480 restartedMain INFO c.z.h.HikariDataSource getConnection 122 H
26-06-26 23:38:08.588 restartedMain WARN o.h.orm.deprecation constructDialect 152 H
26-06-26 23:38:08.628 restartedMain INFO o.h.o.connections.pooling logConnectionInfo 179 H
26-06-26 Database JDBC URL [jdbc:mysql://localhost:3306/ormlearn]
26-06-26 Database driver: MySQL Connector/J
26-06-26 Database dialect: MySQLDialect
26-06-26 Database version: 8.0.45
26-06-26 Default catalog/schema: ormlearn/undefined
26-06-26 Autocommit mode: undefined/unknown
26-06-26 Isolation level: REPEATABLE_READ [default REPEATABLE_READ]
26-06-26 JDBC fetch size: none
26-06-26 Pool: DataSourceConnectionProvider
26-06-26 Minimum pool size: undefined/unknown
26-06-26 Maximum pool size: undefined/unknown
26-06-26 23:38:09.767 restartedMain INFO org.hibernate.orm.core initiateService 53 H
26-06-26 23:38:09.832 restartedMain INFO rEntityManagerFactoryBean buildNativeEntityManagerFactory
26-06-26 23:38:09.966 restartedMain INFO .q.QueryEnhancerFactories <clinit> 48 H
26-06-26 23:38:10.285 restartedMain INFO c.c.o.OrmLearnApplication logStarted 60 S
26-06-26 23:38:10.292 restartedMain INFO c.c.o.OrmLearnApplication main 19 I
26-06-26 23:38:10.292 restartedMain INFO c.c.o.OrmLearnApplication testGetAllCountries 26 S
26-06-26 23:38:10.522 restartedMain DEBUG org.hibernate.SQL logStatement 126 S
26-06-26 23:38:10.566 restartedMain DEBUG c.c.o.OrmLearnApplication testGetAllCountries 31 C
26-06-26 23:38:10.567 restartedMain INFO c.c.o.OrmLearnApplication testGetAllCountries 33 E
26-06-26 23:38:10.580 locationShutdownHook INFO rEntityManagerFactoryBean destroy 693 C
26-06-26 23:38:10.584 locationShutdownHook INFO c.z.h.HikariDataSource close 349 H
26-06-26 23:38:10.599 locationShutdownHook INFO c.z.h.HikariDataSource close 351 H
```

Difference between JPA, Hibernate and Spring Data JPA

Java Persistence API (JPA) is a Java specification for object relational mapping (ORM).It allows you to map Java objects (entities) to database tables and perform CRUD operations without writing raw SQL. JPA requires an implementation such as Hibernate, EclipseLink or OpenJPA to perform actual database operations.

Hibernate is a Java ORM framework that simplifies database operations, including handling queries with IN clauses efficiently .Hibernate is an Object Relational Mapping (ORM) framework for Java that maps Java classes to database tables, eliminating the need for manual JDBC code and improving maintainability and performance. It supports multiple databases through SQL dialects, provides caching mechanisms, and allows object-oriented querying using Hibernate Query Language (HQL).

Spring Data JPA is a Spring framework module that simplifies working with relational databases using the Java Persistence API (JPA).It eliminates most boilerplate code for data access by providing repository interfaces and query generation from method names.